10. If $\frac{\log x}{y-z} = \frac{\log y}{z-x} = \frac{\log z}{x-y}$, then prove that $xyz = 1$.
11. Solve $\log_2 x - 3\log_{\frac{1}{2}} x = 6$.
12. Solve $\log_{5-x}(x^2 - 6x + 65) = 2$.

2.10 Application of Algebra in Real Life

Algebra is used in many aspects of life. Financial planning is an area in daily life where algebra is used. Algebra concepts are used to calculate interest rates by bankers and as well as for calculating loan repayments. They are used to predict growth of money. Physical fitness is another area where calculations are made to determine the right amount of food intake for an individual taking into consideration such as the height, body mass of the person etc. Doctors use algebra in measuring drug dosage depending on age and weight of an individual. Architects depend on algebra to design buildings while civil engineers use it to design roads, bridges and tunnels. Algebra is needed to convert items to scale so that the structures designed have the correct proportions. It is used to programme computers and phones . Let us see some examples. Because of the extra-ordinary range of sensitivity of the human ear (a range of over 1000 million millions to one), it is useful to use logarithmic scale to measure sound intensity over this range. The unit of measure decibel is named after the inventor of the telephone Alexander Graham Bell.
If we know the population in the world today, the growth, which is rapid, can be measured by approximating to an exponential function. The radioactive carbon-14 is an organism which decays according to an exponential formula.

Exercise - 2.13

23CA18

Choose the correct or the most suitable answer.

1. If $|x+2| \leq 9$, then x belongs to
   (1) $(-\infty, -7)$ (2) $[-11, 7]$ (3) $(-\infty, -7) \cup [11, \infty)$ (4) $(-11, 7)$

2. Given that x, y and b are real numbers $x < y, b > 0$, then
   (1) $xb < yb$ (2) $xb > yb$ (3) $xb \leq yb$ (4) $\frac{x}{b} \geq \frac{y}{b}$

3. If $\frac{|x-2|}{x-2} \geq 0$, then x belongs to
   (1) $[2, \infty)$ (2) $(2, \infty)$ (3) $(-\infty, 2)$ (4) $(-2, \infty)$

4. The solution of $5x - 1 < 24$ and $5x + 1 > -24$ is
   (1) $(4, 5)$ (2) $(-5, -4)$ (3) $(-5, 5)$ (4) $(-5, 4)$

5. The solution set of the following inequality $|x-1| \geq |x-3|$ is
   (1) $[0, 2]$ (2) $[2, \infty)$ (3) $(0, 2)$ (4) $(-\infty, 2)$